Moment selection and the Gaussian limit experiment

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This note clarifies the relation between finite-sample generalized moment selection and the Gaussian limit experiment used to study local power and admissibility. The main conclusion is simple. The fixed-drift Gaussian experiment is the correct limit experiment for the question of interest. Standard moment selection retains every inequality whose standardized drift has a finite limit. It deletes a moment only when its slackness diverges on the slower selection scale. Thus moment selection matters at the boundary of the local experiment, not at an ordinary finite point of that experiment.

This distinction also clarifies the multiple-core problem. Deleting an inequality that is far inside the null is different from determining which core moment realizes the maximum in the local mean restriction. The latter cannot be learned without error when the core means have a finite local gap.

1 A convenient normalization

Suppose that the null moment inequalities are

$$E[m_j(W,\theta)] \le 0, \qquad j = 1,\dots,d.$$

Define the studentized statistics and their population drifts by

$$X_{j,n} := \frac{\sqrt{n}\,\overline{m}_{j,n}(\theta_n)}{\widehat{\sigma}_{j,n}}, \qquad \mu_{j,n} := \frac{\sqrt{n}\,E[m_j(W,\theta_n)]}{\sigma_{j,n}}.$$

Positive values indicate violation and negative values indicate slackness. Under the usual local asymptotic approximation,

$$X_n \;\Rightarrow\; X \sim N_d(\mu,\Omega) \qquad \text{when} \qquad \mu_n \to \mu \in \mathbf{R}^d, \tag{1}$$

where Ω is the limiting correlation matrix.

For concreteness, consider the hard-threshold version of generalized moment selection. Let κ_n satisfy

$$\kappa_n \to \infty, \qquad \frac{\kappa_n}{\sqrt{n}} \to 0, \tag{2}$$

and retain moment j when

$$X_{j,n} \ge -\kappa_n.$$

Equivalently, the selected set is

$$\widehat{\mathcal{J}}_n := \{j : X_{j,n} \ge -\kappa_n\}. \tag{3}$$

Other GMS functions lead to the same conclusion below because their argument is X_n/κ_n, which converges to zero under every finite local drift.

2 Two asymptotic scales

The root-n drift $\mu_{j,n}$ governs the limiting statistic and local power. The ratio $\mu_{j,n}/\kappa_n$ governs moment selection. These are different asymptotic objects.

Proposition 1 (What GMS selects)**.** *Suppose that*

$$X_n = \mu_n + Z_n + o_p(1), \qquad Z_n = O_p(1),$$

and let $\widehat{\mathcal{J}}_n$ be defined by (3)*.*

1. *If $\mu_n \to \mu \in \mathbf{R}^d$, then*
$$P(\widehat{\mathcal{J}}_n = \{1, \ldots, d\}) \to 1.$$

2. *If, for some j,*
$$\frac{\mu_{j,n}}{\kappa_n} \to -c$$
with $c > 1$, then moment j is deleted with probability tending to one. If $c < 1$, it is retained with probability tending to one.

3. *If the jth population inequality is separated from equality by a fixed amount, so that*
$$E[m_j(W, \theta_n)] \leq -\delta$$
for some $\delta > 0$ and its standard deviation remains bounded, then moment j is deleted with probability tending to one.

Proof. Since $Z_n = O_p(1)$ and $\kappa_n \to \infty$,

$$\frac{X_n}{\kappa_n} = \frac{\mu_n}{\kappa_n} + o_p(1).$$

If μ_n has a finite limit, every coordinate of this ratio converges to zero, which proves the first claim. Under the second claim, $X_{j,n}/\kappa_n \to_p -c$, so the selection rule is eventually on the appropriate side of -1. For the third claim, $\mu_{j,n}$ is negative and of order $\sqrt{n}$. Condition (2) then gives $\mu_{j,n}/\kappa_n \to -\infty$. □

The boundary case $c = 1$ is deliberately excluded from the second part of the proposition. At that boundary, selection depends on the centering at a finer scale and need not have a unique limit. This is the role of the additional selection-scale parameter in `Addition-2.pdf`.

Proposition 1 distinguishes three uses of the word "slack." A moment with a finite negative value of μ_j is locally slack, but it is statistically close to a binding moment and GMS retains it. A moment with drift of order $-\kappa_n$ lies on the selection scale. A moment that is slack by a fixed population amount has drift of order $-\sqrt{n}$ and is deleted asymptotically.

3 Why the Gaussian experiment does not contain an oracle selector

Moment selection is a component of a testing rule, not a primitive of a statistical experiment. One can define a data-dependent selector in the Gaussian experiment (1), but it cannot identify the binding set without error over the finite mean space.

Proposition 2 (No oracle selection at finite drifts)**.** *Let $X \sim N_d(\mu, \Omega)$ with Ω positive definite. Suppose that two finite mean vectors μ and ν have different binding-set or active-branch labels. There is no measurable selector that returns the correct label with probability one under both $N_d(\mu, \Omega)$ and $N_d(\nu, \Omega)$.*

Proof. Gaussian distributions with a common positive-definite covariance matrix and finite mean vectors are mutually absolutely continuous. Therefore, an event that has probability one under $N_d(\mu, \Omega)$ also has probability one under $N_d(\nu, \Omega)$. If the two correct labels differ, the two events on which a selector returns the correct label are disjoint. They cannot both have probability one. □

This is the limit-experiment version of the contiguity obstruction. Local slackness cannot be consistently distinguished from binding, and two active branches separated by a finite local gap cannot be identified perfectly. The limiting version of standard GMS avoids this classification problem by retaining all finite-drift moments.

4 The two-core problem

Consider the identity-covariance experiment

$$X \sim N_3(\mu, I_3), \qquad \mu_3 \leq \max\{\mu_1, \mu_2\},$$

where coordinates 1 and 2 are core moments and coordinate 3 is redundant. There are two distinct selection questions.

First, GMS asks whether an inequality is sufficiently far inside the null to be deleted. By Proposition 1, both core moments and the redundant moment are retained whenever their local means are finite.

Second, a branchwise gated repair would need to determine which core realizes $\max\{\mu_1, \mu_2\}$. This is not the same question. For example, the natural rule that selects core 1 when $X_1 > X_2$ satisfies

$$P_\mu(X_1 > X_2) = \Phi\left(\frac{\mu_1 - \mu_2}{\sqrt{2}}\right). \tag{4}$$

If core 1 is active, the expression in (4) is the probability of selecting it. This probability equals $1/2$ at a tie and is strictly below one for every finite core gap. It converges to one only when the standardized gap diverges.

Moreover, a core moment can fail to realize the maximum while still having a positive local mean. Such a moment is locally violated, not slack relative to the null, and GMS will not

delete it. Thus ordinary GMS cannot be interpreted as an asymptotically perfect active-core selector.

When one core is slack on the GMS scale, the situation is different. That core is deleted with probability tending to one, and the remaining problem reduces to the one-core experiment. The bivariate gated improvement can then apply to that separated subproblem. This is a lower-dimensional boundary regime, not a solution to the problem over the full finite-drift mean space.

5 Implications for admissibility

Admissibility is defined for a fixed experiment and a fixed parameter space. For local asymptotic admissibility, the relevant experiment is

$$X \sim N_d(\mu, \Omega), \qquad \mu \in \Lambda \subseteq \mathbf{R}^d,$$

with the full vector of finite local means retained. There are three reasons this is the natural object.

1. It is the weak limit of the vector of studentized sample moments along root-n local sequences.
2. It is the fixed-drift limit of the GMS procedure itself: the selection input converges to zero, so the limiting critical value treats every moment as potentially binding.
3. The alternatives for which moment selection deletes a coordinate lie on a different scale. They correspond to sequences escaping to the boundary of the finite mean space rather than to omitted points of the ordinary local experiment.

Consequently, the sensitivity of the long or max test to a finite negative mean is not an artifact. It is exactly the local behavior that the limit experiment is intended to record. Moment selection recovers the appropriate lower-dimensional problem far from the local boundary, but it does not remove the multiple-core adaptation problem within that boundary.

In particular, the unresolved admissibility question in `three-dimensional-case.pdf` remains genuine. A branch-specific bivariate dominator is not a single test that dominates over the union of the two core branches. Any global repair must account for the nonzero probability of selecting the wrong active core when the core gap is finite.

6 A useful scale-separation statement

The connection between the finite-sample and limit analyses can be summarized as follows.

1. On every fixed finite-drift sequence, and uniformly on compact drift sets under the corresponding uniform central limit theorem, GMS retains all moments. The full-dimensional Gaussian test is therefore the relevant object for local power and admissibility.

2. On sequences where an inequality is slack by more than the GMS threshold, that moment is deleted and the limiting testing problem becomes lower-dimensional.

3. Near a multiple-core branch boundary, no selector can determine the active core without nontrivial error. This transition region is where the three-dimensional admissibility problem has bite.

The two-rate analysis in `Addition-2.pdf` supplements the Gaussian limit experiment by describing how the procedure behaves as parameters escape the finite local mean space. It does not replace that experiment. For the admissibility project, the fixed-drift Gaussian experiment is the correct and natural starting point.